LONG-DISTANCE UNIVERSALITY FOR THE BERGER SPHERE AND THE UNICYCLE

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Abstract. Brown, Freedman, Lin and Susskind stated that on the complexity geometry of a single qubit the distance changes by at most $O(\mathcal{I}^{-1/2})$ when the penalty factor $\mathcal{I}$ is sent to infinity, and that parallel parking a unicycle a distance z from the curb costs $z + 2\sqrt{1-\mathcal{I}^{-1}}$ up to terms that vanish at large z. We prove both statements. The largest change of distance on the Berger sphere lies between $\pi/(\sqrt{\mathcal{I}}+\sqrt{\mathcal{I}+1})$ and $\pi/(2\sqrt{\mathcal{I}})$, and the parking cost is $z + 2\sqrt{1-\mathcal{I}^{-1}} + O(1/z)$.

Keywords: complexity geometry; Berger sphere; sub-Riemannian geometry.
MSC 2020: 53C17 (primary); 53C22, 53C30, 81P68.

1. Introduction

Nielsen and his collaborators proposed to measure the difficulty of a quantum computation by the length of the shortest path to the desired unitary in a right-invariant metric that penalizes hard directions [9]. In this *complexity geometry* the unitary group of one qubit, SU(2), carries the metric

$$ds^2 = h_1^2 + h_2^2 + \mathcal{I}\, h_3^2, \qquad \dot{U}U^{-1} = -i\,(h_1\sigma_1 + h_2\sigma_2 + h_3\sigma_3), \tag{1}$$

where σ_j are the Pauli matrices and $\mathcal{I} \geq 1$ is the price of rotating about the third axis. This is a Berger sphere; for $\mathcal{I} = 1$ it is the round sphere of radius one. Brown and Susskind made it the basic example of the subject [6], and Brown, Freedman, Lin and Susskind used it, together with a left-invariant metric on the group of motions of the plane, to argue that complexity geometries with very different short-distance behaviour agree at long distances up to small additive errors [5].

Two quantitative claims of [5] carry this argument. Write $d_{\mathcal{I}}$ for the distance from the identity in (1), and $d_\infty = \lim_{\mathcal{I}\to\infty} d_{\mathcal{I}}$ for the sub-Riemannian distance in which only curves with $h_3 \equiv 0$ are admitted. The first claim, equation (4) of [5], is that

$$d_\infty(U) - d_{\mathcal{I}}(U) \leq O(\mathcal{I}^{-1/2}) \qquad \text{for all } U \in \mathrm{SU}(2) \text{ and } \mathcal{I} \geq 1. \tag{2}$$

It is attributed there to an analysis based on the cut loci of Berger spheres, which have been determined together with the diameter [10]; we have found no proof of (2) in the literature. The second claim, equation (6) of [5], concerns a unicycle that starts parallel to a curb and must end parallel to it after moving sideways by z; with the metric (5) of Section 5, the cost of the best manoeuvre is asserted to be

$$C_{\mathcal{I}}(z) = z + 2\sqrt{1-\mathcal{I}^{-1}} + \ldots, \tag{3}$$

the omitted terms vanishing at large z. The text derives (3) by optimizing the manoeuvre "turn, roll to the curb, turn back". We shall see that this manoeuvre only gives $z + \pi + o(1)$, while (3) is nevertheless correct.

Our first result proves (2) with a constant that is sharp to leading order.

Theorem A. *For every $\mathcal{I} > 0$ and every $U \in \mathrm{SU}(2)$,*

$$0 \le d_\infty(U) - d_{\mathcal{I}}(U) \le \frac{\pi}{2\sqrt{\mathcal{I}}}.$$

Moreover

$$\sup_{U \in \mathrm{SU}(2)} \big(d_\infty(U) - d_{\mathcal{I}}(U)\big) \;\ge\; \frac{\pi}{\sqrt{\mathcal{I}} + \sqrt{\mathcal{I}+1}},$$

so that $\sqrt{\mathcal{I}}\,\sup_U(d_\infty - d_{\mathcal{I}}) \to \pi/2$ as $\mathcal{I} \to \infty$.

The lower bound is attained on the rotations $e^{i\sigma_3 z}$ about the expensive axis, at a distance of order $\mathcal{I}^{-1/2}$ from the identity, which is where [5] located the largest discrepancy. The proof rests on two facts of independent interest: the Berger distance is an infimal convolution of the sub-Riemannian distance with the quadratic cost of the fibre (Proposition 3.1), and the square of the sub-Riemannian distance grows at most linearly, with slope 2π, along each Hopf fibre (Lemma 4.2). The same estimate passes to the complexity of one-qubit states, and we also compute the area of the corresponding deformed Bloch sphere, which grows like $\frac{\pi}{2}\log(4\mathcal{I})$ (Corollary 4.4 and Proposition 4.5); this confirms the remaining claim of the two-dimensional example in [5].

Our second result settles the parking problem.

Theorem B. *Let $1 \le \mathcal{I} \le \infty$, put $\kappa = \sqrt{1 - \mathcal{I}^{-1}}$ (so $\kappa = 1$ when $\mathcal{I} = \infty$), and let $C_{\mathcal{I}}(z)$ be the parking cost defined in Section 5. Then*

$$C_{\mathcal{I}}(z) = z + 2\kappa + O(1/z) \qquad (z \to \infty).$$

More precisely, for every $z > 0$,

$$C_{\mathcal{I}}(z) \;\ge\; z + 2\kappa - \frac{4\kappa(1+\sqrt{\mathcal{I}})}{\kappa z + 2(1+\sqrt{\mathcal{I}})} \quad (\mathcal{I} < \infty), \qquad C_\infty(z) \;\ge\; z + 2 - \frac{4}{z+2},$$

and $C_{\mathcal{I}}(z) \le z + 2\kappa + (2 + 8\kappa + 4\kappa^2)/z$ whenever $1 < \mathcal{I} \le \infty$, $z \ge 2$ and $z \ge (4/\kappa)\log(\pi z)$.

The lower bound comes from two calibrating functions, $z \mp \kappa(1 - \sin\theta)$, whose differentials have unit length everywhere (Lemma 5.1); the upper bound follows their gradient lines, which turn the unicycle while it rolls (Figure 2). Both theorems are proved by direct analytic arguments, and every step is elementary once the classical facts about the sub-Riemannian geometry of SU(2) recalled in Section 2 are granted.

2. The Berger sphere and its sub-Riemannian limit

Throughout, $\mathbb{S}$ denotes the sphere of radius $\frac{1}{2}$ in $\mathbb{R}^3$, of area π and Gaussian curvature 4. We identify $\mathbb{R}^3$ with the traceless Hermitian matrices by $x \mapsto x_1\sigma_1 + x_2\sigma_2 + x_3\sigma_3$, and define

$$\Pi : \mathrm{SU}(2) \to \mathbb{S}, \qquad \Pi(U) = \tfrac{1}{2}\, U^{-1}\sigma_3 U.$$

For a curve $U(t)$ write $A = \dot U U^{-1} = -i\sum_j h_j\sigma_j$ as in (1), and call the curve *horizontal* if $h_3 \equiv 0$. Since $\frac{d}{dt}\Pi(U) = \frac{1}{2}U^{-1}[\sigma_3, A]U$ and $[\sigma_3, A] = 2(h_1\sigma_2 - h_2\sigma_1)$, the base point moves with speed $(h_1^2 + h_2^2)^{1/2}$. Thus Π maps every horizontal curve onto a curve of the same length in $\mathbb{S}$, and the fibres of Π are the circles $\{e^{i\sigma_3 s}U : s \in \mathbb{R}/2\pi\}$. Left multiplication by $e^{i\sigma_3 s}$ preserves (1): it fixes Π, replaces A by $e^{i\sigma_3 s}Ae^{-i\sigma_3 s} + i\dot s\sigma_3$, and conjugation by $e^{i\sigma_3 s}$ rotates the (σ_1, σ_2)-plane and fixes σ_3. We write $p = \Pi(1) = \frac{1}{2}\sigma_3$.

Lemma 2.1 (holonomy)**.** *Let Γ_τ, $\tau \in [\tau_0, \tau_1]$, be a smooth family of curves in $\mathbb{S}$ from p to a fixed point q, let W_τ be the horizontal lift of Γ_τ starting at 1, and let α be the signed area swept by Γ_τ as τ runs from τ_0 to τ_1. Then $W_{\tau_1}(1) = e^{\pm 2i\alpha\sigma_3} W_{\tau_0}(1)$, the sign depending only on the orientation convention for α.*

Proof. Regard h_1, h_2, h_3 as the right-invariant one-forms defined by $dU\,U^{-1} = -i\sum_j h_j\sigma_j$. The Maurer–Cartan equation $d(dU\,U^{-1}) = (dU\,U^{-1}) \wedge (dU\,U^{-1})$ and $[\sigma_1, \sigma_2] = 2i\sigma_3$ give $dh_3 = 2\,h_1 \wedge h_2$. On horizontal vectors $h_1 \wedge h_2$ is the pull-back of the area form of $\mathbb{S}$, because Π is an isometry there. Apply Stokes' theorem to the map $(\tau, t) \mapsto W_\tau(t)$ of the rectangle $[\tau_0, \tau_1] \times [0, 1]$. The form h_3 vanishes on the sides $t = 0$ (a constant point) and $\tau = \tau_0, \tau_1$ (horizontal curves). On the side $t = 1$ the curve $\tau \mapsto W_\tau(1) = e^{i\sigma_3 c(\tau)} W_{\tau_0}(1)$ stays in the fibre over q and $h_3 = -\dot{c}\,d\tau$ there. Hence $\pm(c(\tau_1) - c(\tau_0)) = \int dh_3 = 2\alpha$. □

We use the following facts on the sub-Riemannian structure, due to Boscain and Rossi [4]; see also Montgomery [8, Chapter 1] for the connection with the isoperimetric problem of Dido.[1]

(F1) For $s \in [0, 2\pi]$, $d_\infty(e^{i\sigma_3 s}) = \sqrt{s(2\pi - s)}$.

(F2) Every $V \in \mathrm{SU}(2)$ is joined to 1 by a horizontal curve of length $d_\infty(V)$. Its projection to $\mathbb{S}$ is an arc of a circle (a curve of constant geodesic curvature, possibly a great circle) traversed at most once; when $\Pi(V) \neq p$ it is a proper arc from p to $\Pi(V)$.

The second statement in (F2) holds because a minimizing geodesic stays optimal exactly until its projection closes up, at which moment it meets the fibre over p, the cut locus of [4]. Formula (F1) is the case of a full circle: a circle of $\mathbb{S}$ bounding area $a \in [0, \pi]$ has length $2\sqrt{a(\pi - a)}$, and by Lemma 2.1 its horizontal lift from 1 ends at $e^{\pm 2ia\sigma_3}$; with $s = 2a$ this length is $\sqrt{s(2\pi - s)}$, which is also equation (3) of [5].

3. The Berger distance as an infimal convolution

Proposition 3.1. *For every $\mathcal{I} > 0$ and $U \in \mathrm{SU}(2)$,*

$$d_{\mathcal{I}}(U)^2 = \min_{B \in \mathbb{R}} \Big(d_\infty(e^{i\sigma_3 B} U)^2 + \mathcal{I} B^2 \Big).$$

Proof. Let $U(t)$, $t \in [0, 1]$, join 1 to U, with components $h_j(t)$ as in (1). Put $c(t) = \int_0^t h_3$, $B = c(1)$ and $W(t) = e^{i\sigma_3 c(t)} U(t)$. Then

$$\dot{W} W^{-1} = i\dot{c}\,\sigma_3 + e^{i\sigma_3 c}\,\dot{U} U^{-1} e^{-i\sigma_3 c} = -i\,e^{i\sigma_3 c}(h_1\sigma_1 + h_2\sigma_2)e^{-i\sigma_3 c},$$

so W is horizontal, with horizontal speed $(h_1^2 + h_2^2)^{1/2}$, and it ends at $e^{i\sigma_3 B} U$. The length of $U(t)$ is the Euclidean length of the plane curve with velocity $\big((h_1^2 + h_2^2)^{1/2}, \sqrt{\mathcal{I}}\,h_3\big)$, which is at least the distance between its endpoints:

$$\int_0^1 \sqrt{h_1^2 + h_2^2 + \mathcal{I} h_3^2}\,dt \;\geq\; \Big(\Big(\int_0^1 \sqrt{h_1^2 + h_2^2}\,dt \Big)^2 + \mathcal{I} B^2 \Big)^{1/2} \;\geq\; \Big(d_\infty(e^{i\sigma_3 B} U)^2 + \mathcal{I} B^2 \Big)^{1/2}.$$

Conversely, given B, let W be a horizontal curve of constant speed from 1 to $e^{i\sigma_3 B} U$ of length $d_\infty(e^{i\sigma_3 B} U)$, which exists by (F2), and put $U(t) = e^{-i\sigma_3 B t} W(t)$. The computation above, read backwards, shows that $U(t)$ has constant horizontal speed $d_\infty(e^{i\sigma_3 B} U)$ and $h_3 \equiv B$, so its length is $\big(d_\infty(e^{i\sigma_3 B} U)^2 + \mathcal{I} B^2 \big)^{1/2}$, and it ends at U. The minimum exists since d_∞ is continuous and $\mathcal{I} B^2 \to \infty$. □

[1] In [4] the horizontal metric is four times ours, so their distance is twice d_∞. Their Theorem 2 gives the distance $2\sqrt{\arg\alpha\,(2\pi - \arg\alpha)}$ to $\mathrm{diag}(\alpha, \bar{\alpha}) = e^{i\sigma_3 \arg\alpha}$, and their Theorem 1 identifies the cut locus with the fibre through 1 minus the identity.

Proposition 3.1 makes precise the picture of [5]: the Berger distance spends a part B of the motion along the expensive direction, at quadratic cost $\mathcal{I}B^2$, and synthesizes the rest by commutators of cheap rotations. Since $d_\infty(e^{i\sigma_3 B}U)$ is bounded by the sub-Riemannian diameter, the optimal B tends to 0 as $\mathcal{I} \to \infty$ and $d_{\mathcal{I}} \to d_\infty$ uniformly; Theorem A measures the rate.

4. Proof of Theorem A

4.1. **The pencil of circles through two points.** Fix $q \in \mathbb{S}$ with $q \neq p$. For each unit tangent vector $e_\tau = (\cos\tau)e_1 + (\sin\tau)e_2$ at p there is exactly one circle C_τ of $\mathbb{S}$ tangent to e_τ at p and passing through q; let Γ_τ be the arc of C_τ that starts at p in the direction e_τ and ends at its first arrival at q (Figure 1). Write $f(\tau)$ for the length of Γ_τ, $k(\tau)$ for its constant geodesic curvature, and $\alpha(\tau)$ for the signed area swept by the arcs from Γ_0 to Γ_τ.

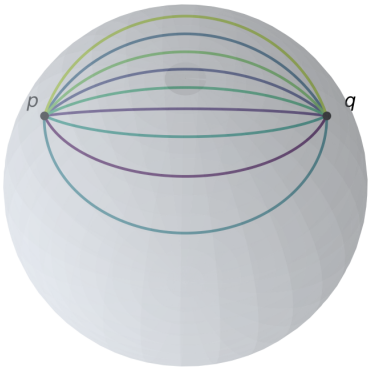


Figure 1. Arcs of the pencil of circles through p and q on the sphere of radius $\frac{1}{2}$. Distinct arcs meet only at p and q, and together they sweep the sphere once.

Lemma 4.1. *The functions f, k, α are smooth, α is strictly increasing with $\alpha(\tau + 2\pi) = \alpha(\tau) + \pi$, and*

$$\Big|\frac{d}{d\tau} f(\tau)^2\Big| \le 4\pi \, \frac{d\alpha}{d\tau}(\tau) \qquad \text{for all } \tau.$$

Proof. Smoothness is clear from the explicit construction. Two distinct circles meet in at most two points, so the arcs Γ_τ have pairwise disjoint interiors; every point of $\mathbb{S} \setminus \{p, q\}$ lies on exactly one circle through p and q, and on exactly one of its two arcs. Hence, as τ increases by 2π, the arcs sweep $\mathbb{S}$ exactly once and monotonically, which gives the statements about α.

Let N_τ be the unit normal of Γ_τ pointing to the side into which the arcs move as τ increases. The first variation of length for a family with fixed endpoints is

$$f'(\tau) = -\int_{\Gamma_\tau} k_N \, \langle \partial_\tau \Gamma, N_\tau \rangle \, ds, \qquad \alpha'(\tau) = \int_{\Gamma_\tau} \langle \partial_\tau \Gamma, N_\tau \rangle \, ds,$$

where k_N is the geodesic curvature measured with respect to N_τ; the tangential part of the variation contributes only boundary terms, which vanish because the endpoints are fixed. Since $k_N = \pm k(\tau)$ is constant along Γ_τ, $|f'(\tau)| = |k(\tau)|\,\alpha'(\tau)$.

It remains to show $|k(\tau)|f(\tau) \le 2\pi$. If C_τ is a great circle, $k = 0$. Otherwise let D be the disc bounded by C_τ on the side towards which it curves. The Gauss–Bonnet formula for D reads $|k|\,L(C_\tau) + 4\,\mathrm{Area}(D) = 2\pi$, so $|k|\,L(C_\tau) \le 2\pi$, and $f(\tau) < L(C_\tau)$ since Γ_τ is a proper arc of C_τ. Therefore $|(f^2)'| = 2f|k|\alpha' \le 4\pi\alpha'$. □

4.2. **Square-root growth along the fibres.**

Lemma 4.2. *For every $V \in \mathrm{SU}(2)$ and $B \in \mathbb{R}$,*

$$d_\infty(e^{i\sigma_3 B}V)^2 \;\le\; d_\infty(V)^2 + 2\pi|B|.$$

Proof. Suppose first $\Pi(V) = q \neq p$. By (F2) a minimizing horizontal curve from 1 to V projects onto an arc Γ_{τ_0} of the pencil, so $f(\tau_0) = d_\infty(V)$. Let W_τ be the horizontal lift of Γ_τ starting at 1. By Lemma 2.1, $W_\tau(1) = e^{i\sigma_3 c(\tau)}V$ with $c(\tau) = \pm 2(\alpha(\tau) - \alpha(\tau_0))$, a fixed sign. By Lemma 4.1, c is a strictly monotone bijection of $\mathbb{R}$, so there is τ_1 with $c(\tau_1) = B$, and $|\alpha(\tau_1) - \alpha(\tau_0)| = |B|/2$. Integrating the inequality of Lemma 4.1 between τ_0 and τ_1,

$$d_\infty(e^{i\sigma_3 B}V)^2 \le f(\tau_1)^2 \le f(\tau_0)^2 + 4\pi|\alpha(\tau_1) - \alpha(\tau_0)| = d_\infty(V)^2 + 2\pi|B|.$$

Suppose next $\Pi(V) = p$, that is $V = e^{i\sigma_3 s}$. Let g be the 2π-periodic function equal to $t(2\pi - t)$ on $[0, 2\pi]$. It is continuous, and $|g'| \le 2\pi$ away from the multiples of 2π, so g is 2π-Lipschitz on $\mathbb{R}$. By (F1), $d_\infty(e^{i\sigma_3 t})^2 = g(t)$, hence $d_\infty(e^{i\sigma_3(s+B)})^2 = g(s+B) \le g(s) + 2\pi|B|$. □

At $V = 1$ the lemma is the familiar square-root law $d_\infty(e^{i\sigma_3 B}) \le \sqrt{2\pi|B|}$ of the ball–box theorem; its content is that the same constant works at every point, with the square of the distance in place of the distance.

4.3. **Conclusion of the proof.**

Proof of Theorem A. Horizontal curves have the same length for every $\mathcal{I}$, so $d_\mathcal{I} \le d_\infty$. Fix U, choose B as in Proposition 3.1, and put $X = d_\infty(e^{i\sigma_3 B}U)$, so that $d_\mathcal{I}(U)^2 = X^2 + \mathcal{I}B^2$. Lemma 4.2, applied to $V = e^{i\sigma_3 B}U$ and the shift $-B$, gives $d_\infty(U)^2 \le X^2 + 2\pi|B|$. If $2\pi|B| \le \mathcal{I}B^2$ then $d_\infty(U) \le d_\mathcal{I}(U)$. Otherwise, since $x \mapsto \sqrt{x+a} - \sqrt{x+b}$ decreases on $[0,\infty)$ when $a > b \ge 0$,

$$\begin{aligned} d_\infty(U) - d_\mathcal{I}(U) &\le \sqrt{X^2 + 2\pi|B|} - \sqrt{X^2 + \mathcal{I}B^2} \le \sqrt{2\pi}\,t - \sqrt{\mathcal{I}}\,t^2\Big|_{t=\sqrt{|B|}} \\ &\le \max_{t\ge 0}\bigl(\sqrt{2\pi}\,t - \sqrt{\mathcal{I}}\,t^2\bigr) = \frac{\pi}{2\sqrt{\mathcal{I}}}. \end{aligned}$$

For the lower bound take $U = e^{i\sigma_3 z}$ with $z \in (0, \pi]$. The curve $e^{i\sigma_3 tz}$, $t \in [0,1]$, has $h_3 \equiv -z$ and length $\sqrt{\mathcal{I}}\,z$, so $d_\mathcal{I}(U) \le \sqrt{\mathcal{I}}z$, while $d_\infty(U) = \sqrt{z(2\pi - z)}$ by (F1). Choose $z = \pi\bigl(1 - \sqrt{\mathcal{I}/(\mathcal{I}+1)}\bigr)$. Then $2\pi z - z^2 = \pi^2/(\mathcal{I}+1)$ and

$$d_\infty(U) - d_\mathcal{I}(U) \;\ge\; \frac{\pi}{\sqrt{\mathcal{I}+1}} - \sqrt{\mathcal{I}}\,\pi\Bigl(1 - \sqrt{\tfrac{\mathcal{I}}{\mathcal{I}+1}}\Bigr) = \pi\bigl(\sqrt{\mathcal{I}+1} - \sqrt{\mathcal{I}}\bigr) = \frac{\pi}{\sqrt{\mathcal{I}} + \sqrt{\mathcal{I}+1}}. \quad \square$$

Remark 4.3. The point $e^{i\sigma_3 z}$ used for the lower bound has $d_\mathcal{I} \le \sqrt{\mathcal{I}}\,z \approx \pi/(2\sqrt{\mathcal{I}})$, a curvature length from the identity, in agreement with the location of the largest discrepancy described in [5]. In the normalization (1) the direct synthesis of $e^{i\sigma_3 z}$ costs $\sqrt{\mathcal{I}}\,z$; the value $\mathcal{I}z$ printed in [5] corresponds to measuring the penalty in units of $\sqrt{\mathcal{I}}$ and does not affect any conclusion there. The two bounds of Theorem A differ by $O(\mathcal{I}^{-3/2})$. Whether the supremum equals

$\pi/(\sqrt{\mathcal{I}}+\sqrt{\mathcal{I}+1})$ for every $\mathcal{I}$, that is, whether it is always attained on the fibre through the identity, is left open.

4.4. **One-qubit states and the deformed Bloch sphere.** The complexity of a state ψ relative to a reference state ψ_0 is the least complexity of a unitary carrying ψ_0 to ψ [6]:

$$\delta_{\mathcal{I}}(\psi_0,\psi)=\min\{d_{\mathcal{I}}(U):U\psi_0=\psi\}.$$

Corollary 4.4. *For all states ψ_0,ψ of one qubit and every $\mathcal{I}>0$, $0\le\delta_\infty(\psi_0,\psi)-\delta_{\mathcal{I}}(\psi_0,\psi)\le\pi/(2\sqrt{\mathcal{I}})$.*

Proof. Both minima are over the same closed set of unitaries, on which $d_{\mathcal{I}}\le d_\infty\le d_{\mathcal{I}}+\pi/(2\sqrt{\mathcal{I}})$ by Theorem A. □

When ψ_0 is the north pole of the Bloch sphere, Brown and Susskind showed that $\delta_{\mathcal{I}}$ is the distance of the metric

$$4\,ds^2=d\theta^2+\frac{\mathcal{I}\sin^2\theta}{\mathcal{I}\cos^2\theta+\sin^2\theta}\,d\phi^2 \tag{4}$$

on the two-sphere [6, equations (3.9)–(3.10)]. By Corollary 4.4, sending $\mathcal{I}\to\infty$ changes all its distances by less than $\pi/(2\sqrt{\mathcal{I}})$. The area behaves very differently.

Proposition 4.5. *For $\mathcal{I}>1$ the area of (4) equals*

$$\pi\sqrt{\frac{\mathcal{I}}{\mathcal{I}-1}}\ \operatorname{arsinh}\sqrt{\mathcal{I}-1}\ =\ \frac{\pi}{2}\log(4\mathcal{I})+o(1)\qquad(\mathcal{I}\to\infty).$$

Proof. The area element of (4) is $\frac14\sqrt{\mathcal{I}}\sin\theta\,(\mathcal{I}\cos^2\theta+\sin^2\theta)^{-1/2}d\theta\,d\phi$. Integrating over ϕ and substituting $u=\cos\theta$,

$$\text{Area}=\frac{\pi}{2}\sqrt{\mathcal{I}}\int_{-1}^{1}\frac{du}{\sqrt{1+(\mathcal{I}-1)u^2}}=\frac{\pi\sqrt{\mathcal{I}}}{\sqrt{\mathcal{I}-1}}\operatorname{arsinh}\sqrt{\mathcal{I}-1}.$$

Since $\operatorname{arsinh}x=\log(x+\sqrt{1+x^2})=\log(2x)+O(x^{-2})$, the area is $\frac{\pi}{2}\log(4\mathcal{I})+o(1)$. □

For $\mathcal{I}=1$ the formula, read as a limit, gives the area π of the round Bloch sphere of radius $\frac12$. Proposition 4.5 confirms the statement of [5, Supplementary Information 3] that the volume diverges logarithmically while distances move by a vanishing amount.

5. Parallel parking

5.1. **The metric and the calibrating functions.** The configuration of the unicycle is its position (y,z) in the plane, with z the distance towards the curb, and its heading $\theta\in\mathbb{R}/2\pi$, the rolling direction being $(\cos\theta,\sin\theta)$. Following [5] we use

$$ds^2=(\cos\theta\,dy+\sin\theta\,dz)^2+\mathcal{I}\,(\sin\theta\,dy-\cos\theta\,dz)^2+d\theta^2,\qquad 1\le\mathcal{I}<\infty, \tag{5}$$

and for $\mathcal{I}=\infty$ the sub-Riemannian metric obtained by admitting only curves with $\sin\theta\,dy-\cos\theta\,dz=0$ (the unicycle cannot drift). In terms of the rolling speed u, the drift speed v and the turning speed $w=\dot\theta$,

$$\dot y=u\cos\theta+v\sin\theta,\qquad \dot z=u\sin\theta-v\cos\theta,\qquad ds^2=(u^2+\mathcal{I}v^2+w^2)\,dt^2.$$

The parking cost $C_{\mathcal{I}}(z)$ is the infimum of the lengths of curves from $(0,0,0)$ to $(0,z,0)$; the metric (5) is invariant under rigid motions of the plane, which is the left-invariance of [5] in the group of motions of the plane, studied in the sub-Riemannian case by Sachkov [11]. Since $(\cos\theta\,dy+\sin\theta\,dz)^2+(\sin\theta\,dy-\cos\theta\,dz)^2=dy^2+dz^2$ and $\mathcal{I}\ge1$, every curve is at least as

long as its projection to the plane, so $C_{\mathcal{I}}(z) \geq z$; for $\mathcal{I} = 1$ drifting straight to the curb shows $C_1(z) = z$. From now on $\mathcal{I} > 1$.

For a function $\varphi(y, z, \theta)$ and a curve of unit speed, Cauchy–Schwarz gives $|\dot\varphi| \leq \|d\varphi\|$, where for $d\varphi = a\,dy + b\,dz + c\,d\theta$

$$\|d\varphi\|^2 = (a\cos\theta + b\sin\theta)^2 + \mathcal{I}^{-1}(a\sin\theta - b\cos\theta)^2 + c^2,$$

with $\mathcal{I}^{-1} = 0$ when $\mathcal{I} = \infty$. If $\|d\varphi\| \leq 1$ everywhere, every curve from P to Q has length at least $|\varphi(Q) - \varphi(P)|$.

Lemma 5.1. *Let $\kappa = \sqrt{1 - \mathcal{I}^{-1}}$, $\beta(\theta) = \kappa(1 - \sin\theta)$ and $\tilde\beta(\theta) = \kappa(1 + \sin\theta)$. Each of the four functions $z \pm \beta(\theta)$, $z \pm \tilde\beta(\theta)$ has $\|d\varphi\| = 1$ at every point.*

Proof. Here $a = 0$, $b = 1$ and $c = \pm\kappa\cos\theta$, so $\|d\varphi\|^2 = \sin^2\theta + \mathcal{I}^{-1}\cos^2\theta + (1 - \mathcal{I}^{-1})\cos^2\theta = 1$. □

5.2. **The lower bound.** Let γ be a curve of length L from $(0, 0, 0)$ to $(0, z, 0)$, and let $M = \max_\gamma |\sin\theta|$, attained at a point $m = (y_m, z_m, \theta_m)$. Suppose $\sin\theta_m = M$. Using $\varphi_- = z - \beta$ on the part of γ before m and $\varphi_+ = z + \beta$ on the part after m,

$$L \;\geq\; \big[z_m - \kappa(1 - M) + \kappa\big] + \big[z + \kappa - z_m - \kappa(1 - M)\big] = z + 2\kappa M.$$

If $\sin\theta_m = -M$, the same computation with $z \mp \tilde\beta$ gives the same bound. On the other hand $|\dot z| \leq \|dz\| = (\sin^2\theta + \mathcal{I}^{-1}\cos^2\theta)^{1/2} \leq (M^2 + \mathcal{I}^{-1}(1 - M^2))^{1/2}$ along γ, so

$$L \;\geq\; z\,r(M), \qquad r(M) = \big(M^2 + \mathcal{I}^{-1}(1 - M^2)\big)^{-1/2}.$$

Write $\varepsilon = 1 - M$. For $\mathcal{I} < \infty$, $r^2 - 1 = \kappa^2(1 - M^2)/(1 - \kappa^2(1 - M^2)) \geq \kappa^2\varepsilon$ and $1 \leq r \leq \sqrt{\mathcal{I}}$, so $r - 1 \geq \kappa^2\varepsilon/(1 + \sqrt{\mathcal{I}})$; for $\mathcal{I} = \infty$, $r - 1 = \varepsilon/(1 - \varepsilon) \geq \varepsilon$. In both cases $L - z \geq \max\{a(1 - \varepsilon),\, b\varepsilon\} \geq ab/(a + b)$ with $a = 2\kappa$ and $b = z\kappa^2/(1 + \sqrt{\mathcal{I}})$, respectively $a = 2$, $b = z$. This is the lower bound of Theorem B.

5.3. **The upper bound.** The construction follows the gradient lines of the calibrating functions. On a unit-speed curve, $\dot\varphi_- = 1$ for $\varphi_- = z - \beta$ forces

$$u = \sin\theta, \qquad v = -\mathcal{I}^{-1}\cos\theta, \qquad w = \kappa\cos\theta, \tag{6}$$

and then $\dot z = \sin^2\theta + \mathcal{I}^{-1}\cos^2\theta$ and $\dot y = \kappa^2\sin\theta\cos\theta$. For $\mathcal{I} = \infty$ the drift vanishes, as it must. Fix $\varepsilon \in (0, \pi/2)$ and put $\theta_1 = \pi/2 - \varepsilon$. The manoeuvre has five stages (Figure 2).

(1) Follow (6) from heading 0 to heading θ_1. Since φ_- increases at unit rate, the cost equals the increase of z plus $\beta(0) - \beta(\theta_1) = \kappa\cos\varepsilon$. With $ds = d\theta/(\kappa\cos\theta)$, the gain in z is

$$Z_1 = \int_0^{\theta_1} \frac{\sin^2\theta + \mathcal{I}^{-1}\cos^2\theta}{\kappa\cos\theta}\,d\theta \leq \frac{1}{\kappa}\log\frac{1 + \cos\varepsilon}{\sin\varepsilon} \leq \frac{1}{\kappa}\log\frac{\pi}{\varepsilon},$$

and the gain in y is $\kappa\int_0^{\theta_1}\sin\theta\,d\theta = \kappa(1 - \sin\varepsilon)$.

(2) Turn on the spot from θ_1 to $\pi/2$, at cost ε.

(3) With $Y = 2\kappa(1 - \sin\varepsilon)$, $m = z - 2Z_1 > 0$ and $\delta = \arctan(Y/m)$, turn to heading $\pi/2 + \delta$, roll a length $\sqrt{m^2 + Y^2}$, and turn back to $\pi/2$. This moves the unicycle by $(-Y, m)$ at cost $\sqrt{m^2 + Y^2} + 2\delta \leq m + (Y^2/2 + 2Y)/m$.

(4) Turn on the spot from $\pi/2$ to θ_1, at cost ε.

(5) Follow $u = \sin\theta$, $v = -\mathcal{I}^{-1}\cos\theta$, $w = -\kappa\cos\theta$ from θ_1 down to 0. This is the gradient line of $\varphi_+ = z + \beta$, so again the cost is the gain in z, equal to Z_1, plus $\kappa\cos\varepsilon$, and the gain in y is again $\kappa(1 - \sin\varepsilon)$.

The unicycle ends at $(0, z, 0)$, and the total cost is at most

$$z + 2\kappa\cos\varepsilon + 2\varepsilon + \frac{2\kappa^2 + 4\kappa}{m}.$$

Take $\varepsilon = 1/z$. If $z \geq 2$ and $z \geq (4/\kappa)\log(\pi z)$, then $2Z_1 \leq (2/\kappa)\log(\pi z) \leq z/2$, so $m \geq z/2$ and the cost is at most $z + 2\kappa + (2 + 8\kappa + 4\kappa^2)/z$. This completes the proof of Theorem B. □

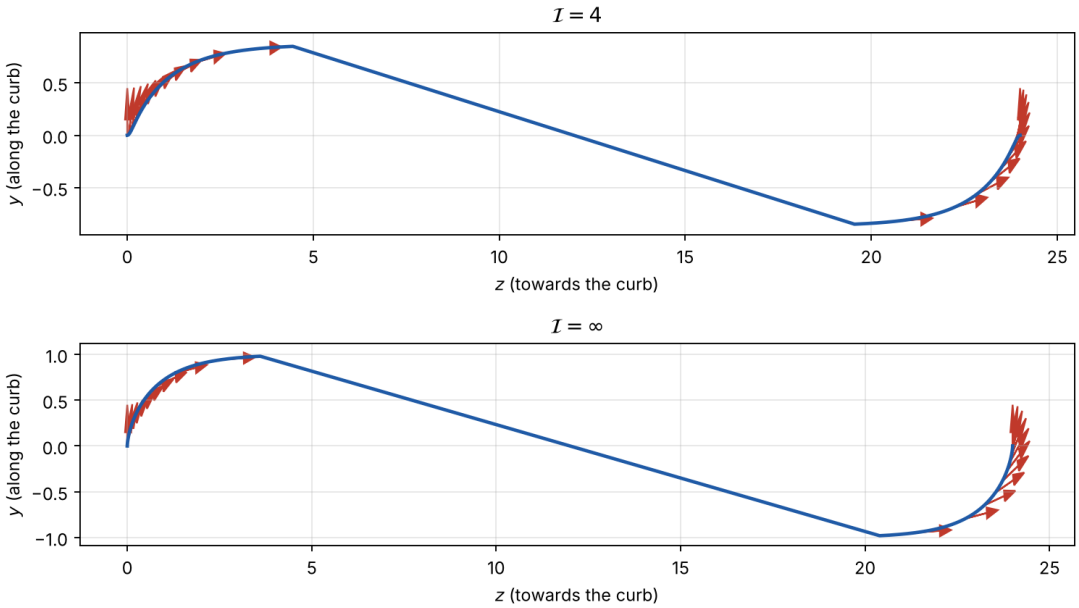


FIGURE 2. The manoeuvre of the upper bound for $z = 24$, for $\mathcal{I} = 4$ (top) and $\mathcal{I} = \infty$ (bottom). Arrows show the heading. The unicycle turns while it rolls, drifts back across the long middle stage, and turns back while it rolls into the parking spot.

5.4. **The manoeuvre turn, roll, turn.** The upper bound $C_{\mathcal{I}}(z) \leq z + \pi$ of [5], obtained by turning through a right angle, rolling to the curb and turning back, holds for every z. Optimizing over the turning angle ϕ, with the middle stage replaced by the straight motion that combines rolling and drifting, costs $2\phi + z\,n(\phi)$, $n(\phi) = (\sin^2\phi + \mathcal{I}\cos^2\phi)^{1/2}$. With $c = \cos\phi$ one has $2\phi \geq \pi - \pi c$ and $n(\phi) \geq 1 + (\mathcal{I} - 1)c^2/(2\sqrt{\mathcal{I}})$, so

$$2\phi + z\,n(\phi) - z \;\geq\; \pi - \pi c + \frac{z(\mathcal{I} - 1)c^2}{2\sqrt{\mathcal{I}}} \;\geq\; \pi - \frac{\pi^2\sqrt{\mathcal{I}}}{2(\mathcal{I} - 1)z}.$$

Hence the optimized turn–roll–turn manoeuvre costs $z + \pi + o(1)$ for every $\mathcal{I} > 1$, strictly more than the true cost $z + 2\kappa + O(1/z)$, since $2\kappa < 2 < \pi$. The constant $2\sqrt{1 - \mathcal{I}^{-1}}$ of (3) is correct, but it is reached only by turning and rolling at the same time, as in (6). For $\mathcal{I} = 1$ the two constants are 0 and π, and the optimum is not to turn at all.

6. Discussion

Theorems A and B make the two low-dimensional examples of [5] rigorous. On the Berger sphere the volume tends to infinity with $\mathcal{I}$, while by Theorem A no distance moves by more than $\pi/(2\sqrt{\mathcal{I}})$. For the unicycle the short-distance geometry depends strongly on $\mathcal{I}$, but by

Theorem B the long-distance growth has slope 1 for every $\mathcal{I} \geq 1$, and $\mathcal{I}$ enters only through the additive constant $2\sqrt{1-\mathcal{I}^{-1}}$, a smooth function of $\mathcal{I}^{-1}$ on $[0,1]$. In the language of [5] the penalty factor is an irrelevant deformation, and Theorem B identifies its effect exactly. The two mechanisms behind the proofs, a quadratic infimal convolution along a central direction and calibration by functions with unit differential, are not special to these groups, and it would be interesting to know how far they reach towards the high-dimensional conjectures of [5].

Acknowledgements

The author acknowledges that the questions treated here were formulated by Brown, Freedman, Lin and Susskind [5], building on the single-qubit analysis of Brown and Susskind [6] and on the complexity geometry of Nielsen, Dowling, Gu and Doherty [9]; the sub-Riemannian facts used in Section 2 are due to Boscain and Rossi [4]. The related negative-curvature picture of complexity is that of Brown, Susskind and Zhao [7]. Earlier work of the author on hyperbolic groups, hypercomplex structures and toric geometry appears in [1, 2, 3].

Declarations

Data availability. No data were generated. The LaTeX source and the script that draws the figures are archived at doi:10.5281/zenodo.24126945.

Funding. The author received no funding for this work.

Conflict of interest. The author declares no conflict of interest.

Ethics statement. This work involves no human or animal subjects.

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